

《数据库设计说明书》- 数据库概述

数据库类型: MySQL (关系型数据库)

设计思路:

本系统数据库设计遵循第三范式 (3NF), 以用户为核心实体, 围绕作文的提交与润色流程建立数据关联。核心实体包括用户 (users)、作文 (essays)、润色记录 (polish\_records) 和润色结果 (polish\_results), 实体之间通过外键建立一对多或一对一的关联关系, 确保数据完整性和查询效率。

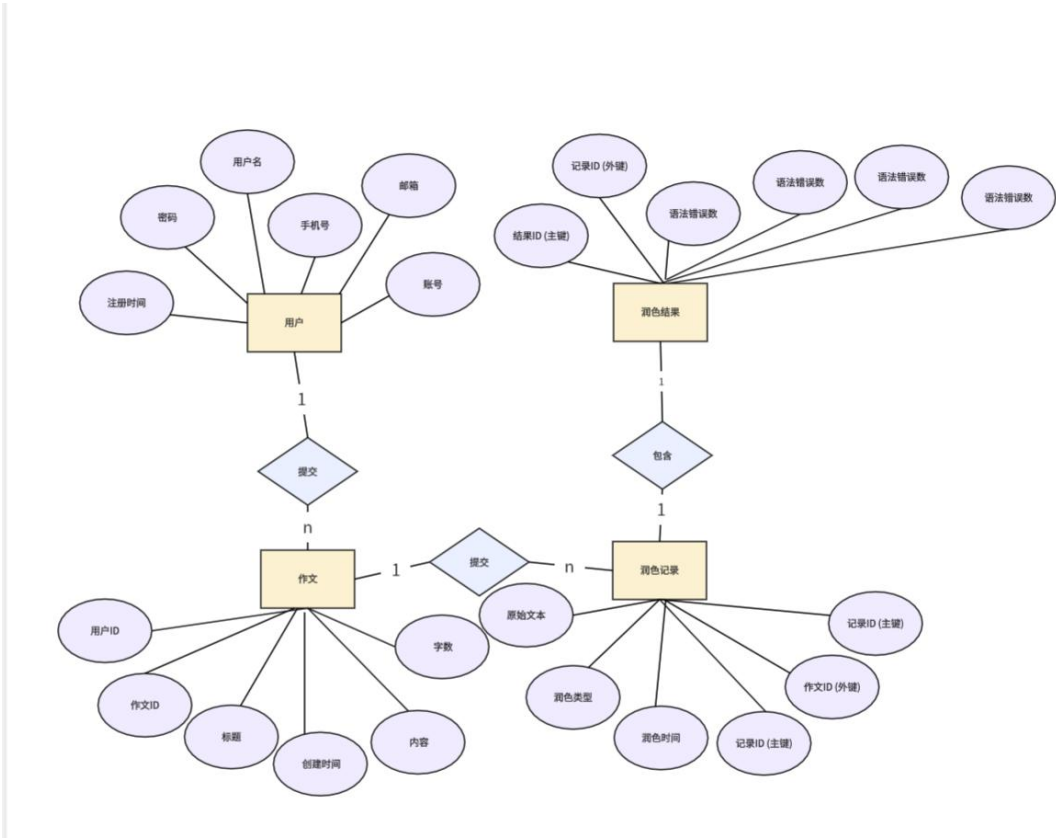
核心表说明:

- users: 存储用户账号信息, 包括用户名、密码、邮箱等
- essays: 存储用户提交的作文, 外键关联用户 ID
- polish\_records: 存储每次润色的记录, 外键关联作文 ID
- polish\_results: 存储润色的详细结果数据, 外键关联润色记录 ID

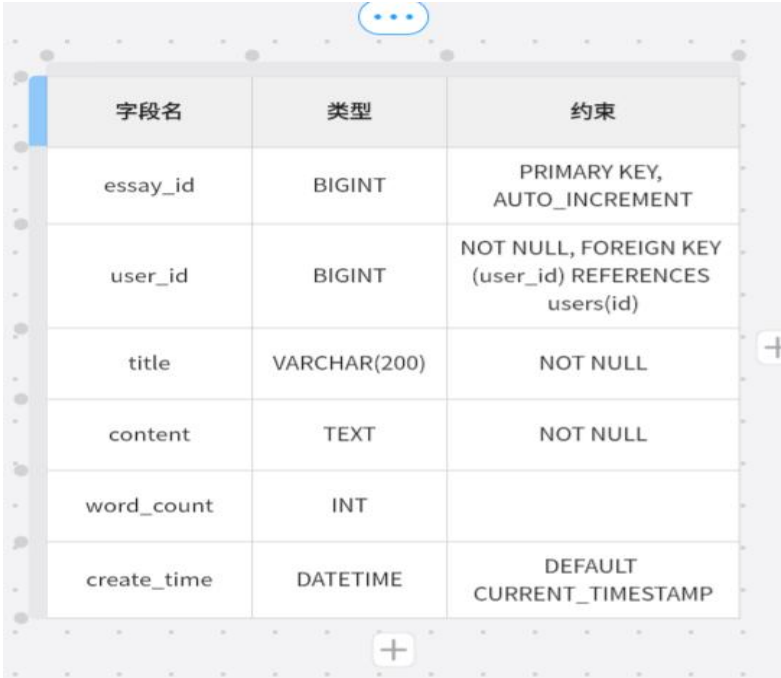
关系概览:

一个用户可提交多篇作文 (1:N), 一篇作文可进行多次润色 (1:N), 一次润色对应一份润色结果 (1:1)

E-R 图



作文表

The image shows a table design tool interface. At the top, there is a blue circle with three dots. The table itself has a light gray header with three columns: '字段名' (Field Name), '类型' (Type), and '约束' (Constraint). The table is surrounded by a gray border with small dots. On the right side of the table, there is a plus sign icon. At the bottom center, there is a plus sign icon inside a circle.

| 字段名         | 类型           | 约束                                                   |
|-------------|--------------|------------------------------------------------------|
| essay_id    | BIGINT       | PRIMARY KEY, AUTO_INCREMENT                          |
| user_id     | BIGINT       | NOT NULL, FOREIGN KEY (user_id) REFERENCES users(id) |
| title       | VARCHAR(200) | NOT NULL                                             |
| content     | TEXT         | NOT NULL                                             |
| word_count  | INT          |                                                      |
| create_time | DATETIME     | DEFAULT CURRENT_TIMESTAMP                            |

润色记录表

| 字段名           | 类型           | 约束                                                                 |
|---------------|--------------|--------------------------------------------------------------------|
| record_id     | BIGINT       | PRIMARY KEY,<br>AUTO_INCREMENT                                     |
| essay_id      | BIGINT       | NOT NULL, FOREIGN KEY<br>(essay_id) REFERENCES<br>essays(essay_id) |
| original_text | TEXT         | NOT NULL                                                           |
| polished_text | TEXT         |                                                                    |
| polish_type   | VARCHAR(50)  |                                                                    |
| score         | DECIMAL(3,2) |                                                                    |
| polish_time   | DATETIME     | DEFAULT<br>CURRENT_TIMESTAMP                                       |

润色结果表

| 字段名                         | 类型           | 约束                                                                           |
|-----------------------------|--------------|------------------------------------------------------------------------------|
| result_id                   | BIGINT       | PRIMARY KEY,<br>AUTO_INCREMENT                                               |
| record_id                   | BIGINT       | NOT NULL, FOREIGN KEY<br>(record_id) REFERENCES<br>polish_records(record_id) |
| grammar_error_count         | INT          |                                                                              |
| vocabulary_suggestion_count | INT          |                                                                              |
| style_score                 | DECIMAL(3,2) |                                                                              |
| overall_comment             | TEXT         |                                                                              |

对象关系映射表

| 实体类          | 对应数据库表         | 说明                          |
|--------------|----------------|-----------------------------|
| User         | users          | 用户实体存储账号信息                  |
| Essay        | essays         | 作文实体，外键关联users              |
| PolishRecord | polish_records | 润色记录，外键关联essays             |
| PolishResult | polish_results | 润色结果，外键关联<br>polish_records |